Hes·so VALAIS WALLIS

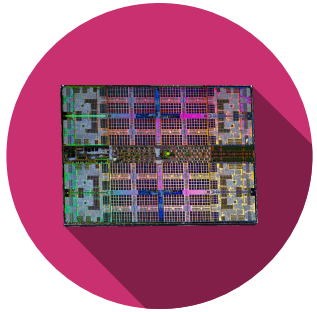
School of Engineering

Computer Architecture

# Micro-Architecture

Arc

Information and Communication Systems program

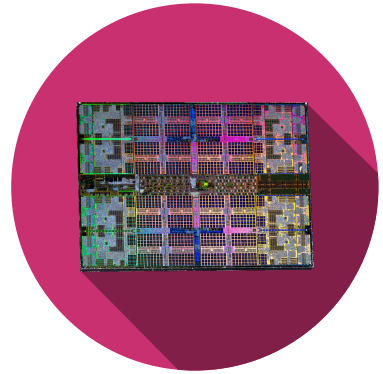


Silvan Zahno silvan.zahno@hevs.ch

# Recapitulation: CPU Performance Equation

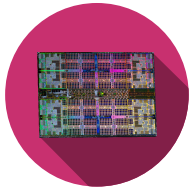
$$T = IC * CPI * CT = \frac{IC * CPI}{f}$$

- $T = \textit{Execution time}$
- $IC = N_{instr}$ = # instructions executed (**I**nstruction **C**ount)
- $CPI$ = **C**ycles **P**er **I**nstruction
- $CT = t_{cycle}$ = **C**ycle **T**ime = duration of clock cycle
- $f$ = clock frequency = $\frac{1}{t_{cycle}}$



# Architectures vs. Microarchitectures

# Architecture vs. Microarchitecture



- "Architecture"/Instruction Set Architecture:
  - Programmer visible state (Memory & Register) • Operations (Instructions and how they work)
  - Execution Semantics (interrupts)
  - Input/Output
  - Data Types/Sizes
- Microarchitecture/Organization:
  - Tradeoffs on how to implement ISA for some metric (Speed, Energy, Cost)
  - Examples: Pipeline depth, number of pipelines, cache size, silicon area, peak power, execution ordering, bus widths, ALU widths

# Architecture vs. Microarchitecture

## Same Architecture / Different Microarchitecture

### AMD Phenom X4

- x86 Instruction Set
- Quad Core
- 125W
- Decode 3 Instructions/Cycle/Core
- 64KB L1 I Cache, 64KB L1 D Cache
- 512KB L2 Cache
- Out-of-order
- 2.6GHz

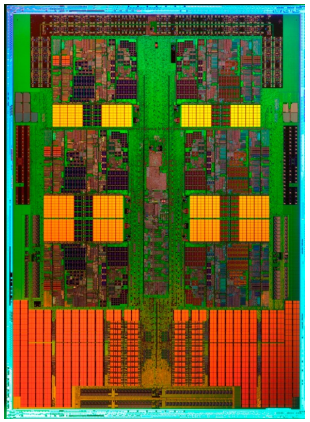


Image Credit: AMD

### Intel Atom

- x86 Instruction Set
- Single Core
- 2W
- Decode 2 Instructions/Cycle/Core
- 32KB L1 I Cache, 24KB L1 D Cache
- 512KB L2 Cache
- In-order
- 1.6GHz

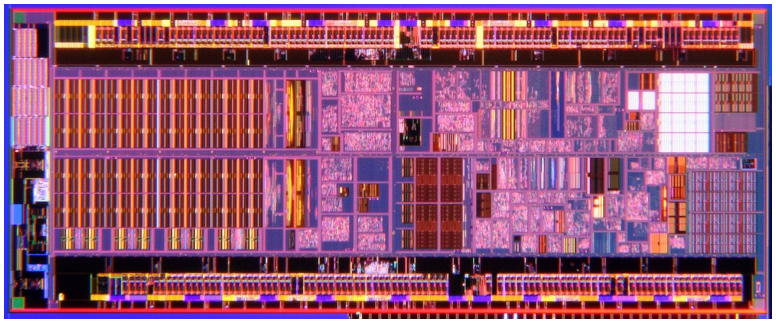


Image Credit: Intel

# Architecture vs. Microarchitecture

## Different Architecture / Different Microarchitecture

### AMD Phenom X4

- X86 Instruction Set
- Quad Core
- 125W
- Decode 3 Instructions/Cycle/Core
- 64KB L1 I Cache, 64KB L1 D Cache
- 512KB L2 Cache
- Out-of-order
- 2.6GHz

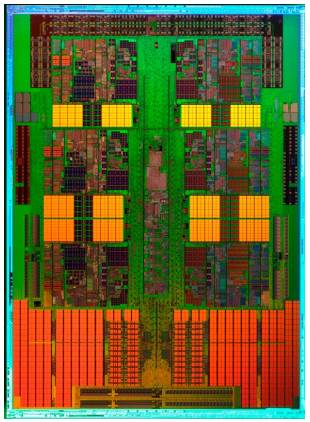


Image Credit: AMD

### IBM POWER7

- Power Instruction Set
- Eight Core
- 200W
- Decode 6 Instructions/Cycle/Core
- 32KB L1 I Cache, 32KB L1 D Cache
- 256KB L2 Cache
- Out-of-order
- 4.25GHz

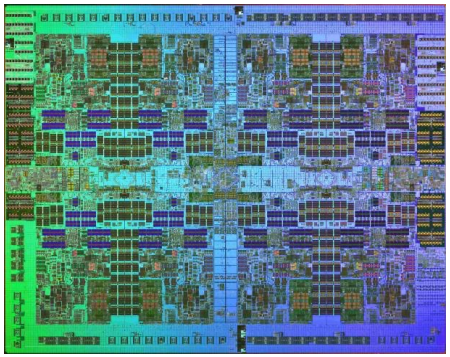
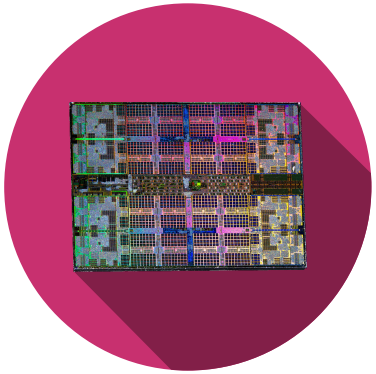


Image Credit: IBM



# Machine Models

# Machine Models

## Elements

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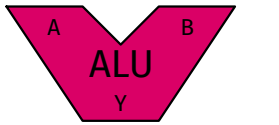


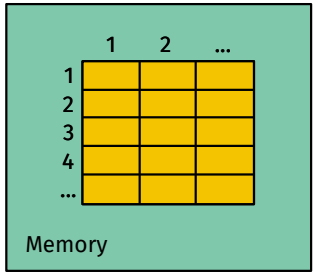

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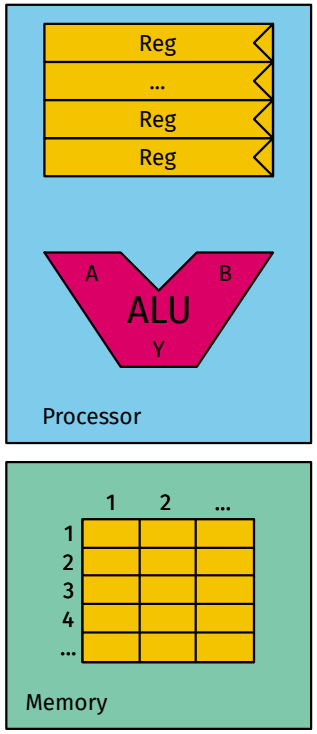
## Elements



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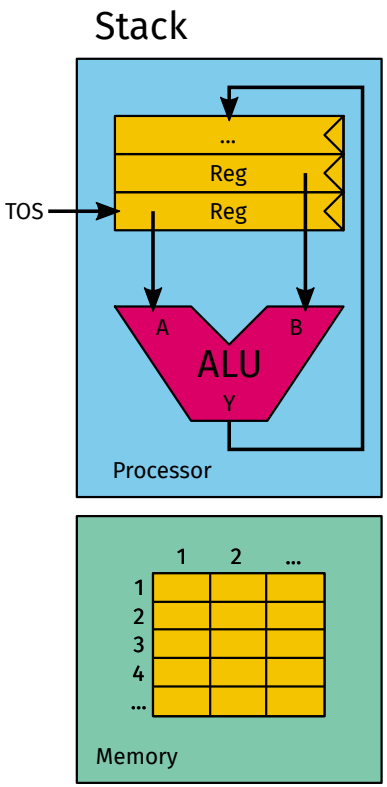
## Elements



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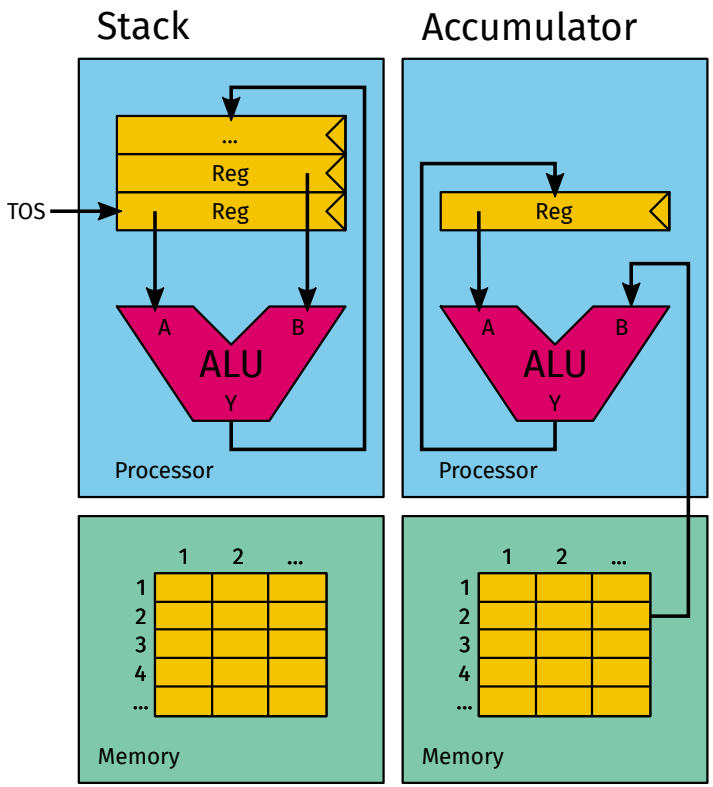

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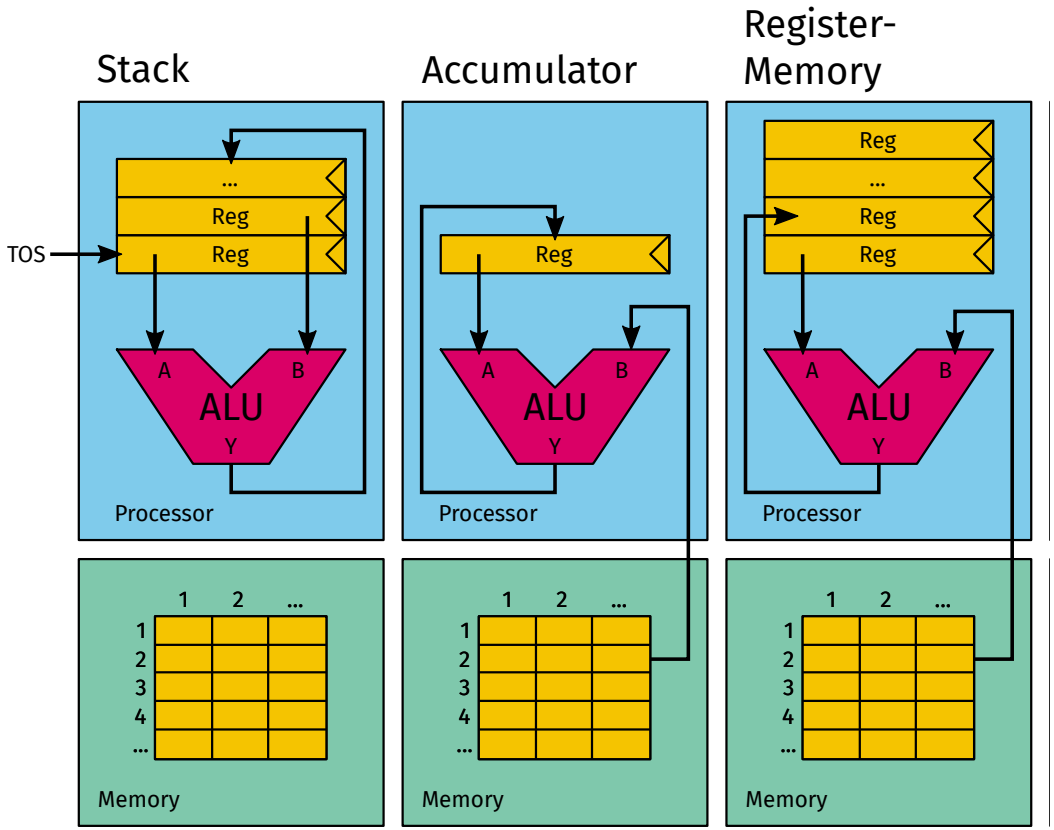

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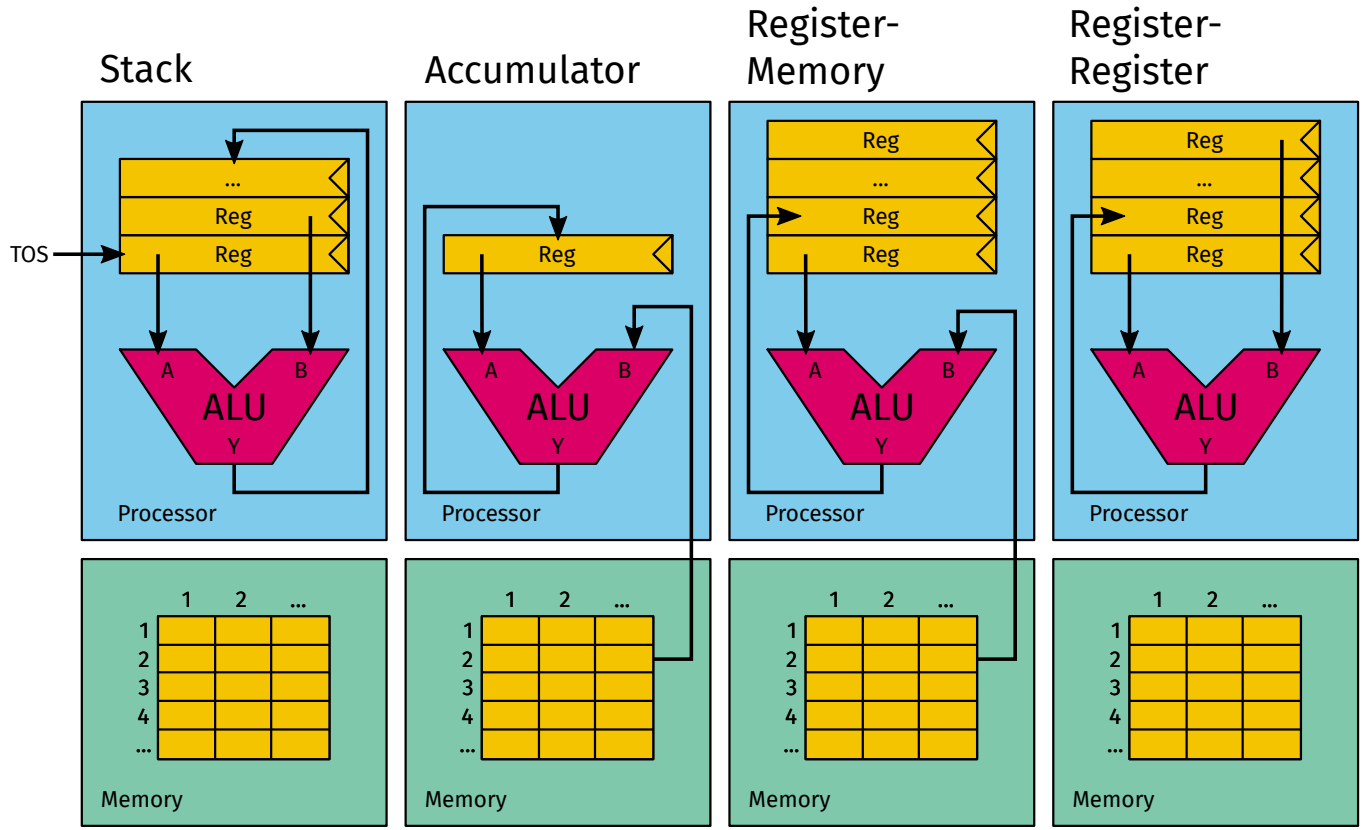

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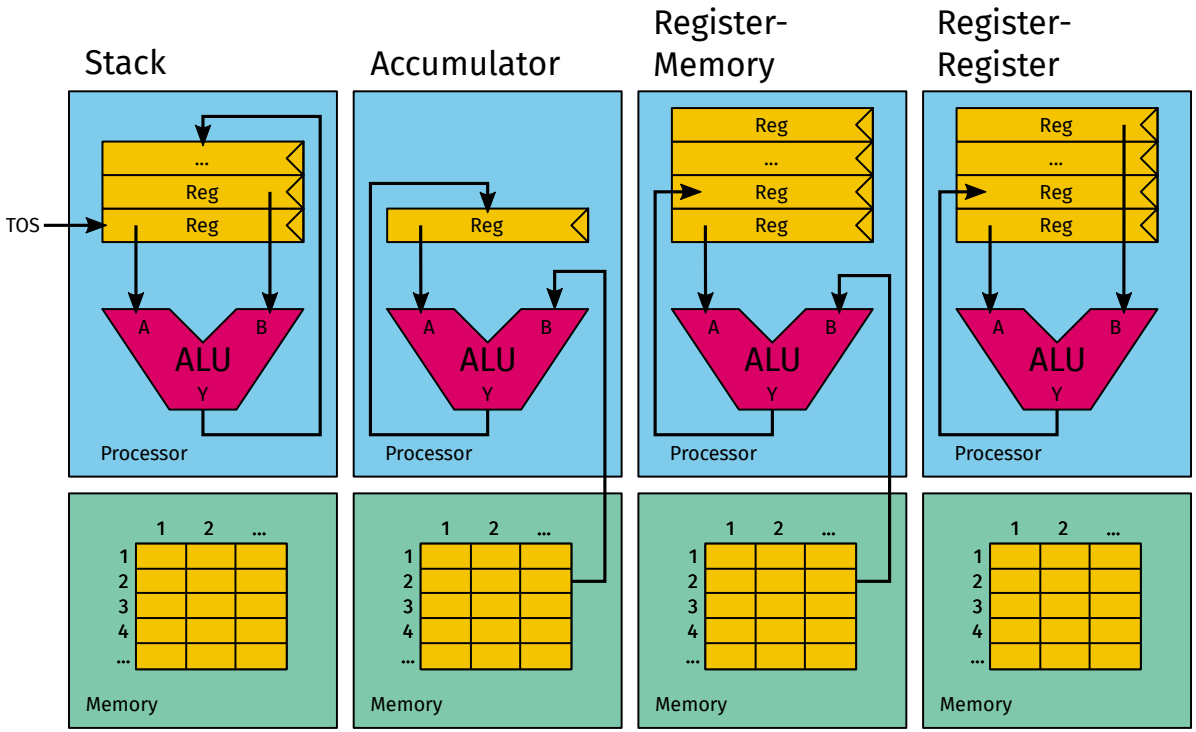
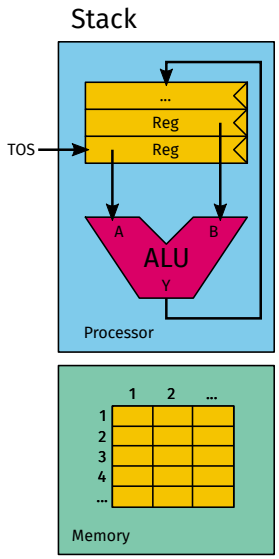



| | Stack | Accumulator | Register-Memory | Register-Register |
|---|---|---|---|---|
| Number Explicitly Named Operands: | 0 | 1 | 2 or 3 | 2 or 3 |

# Machine Models

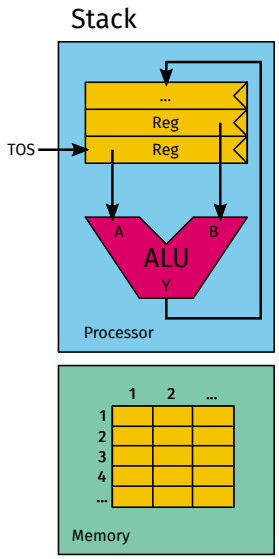
## Example Stack machine model

(a+b*c)/(a+d*c-e)



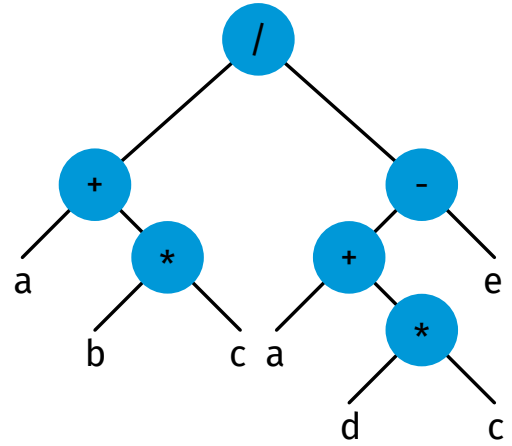
# Machine Models

## Example Stack machine model



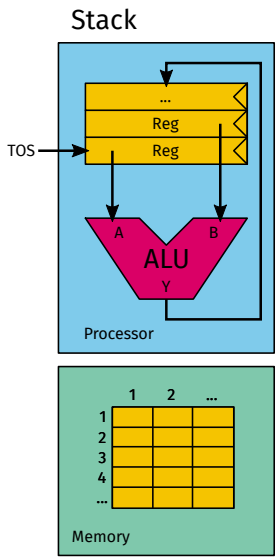

(a+b*c)/(a+d*c-e)

Parsetree



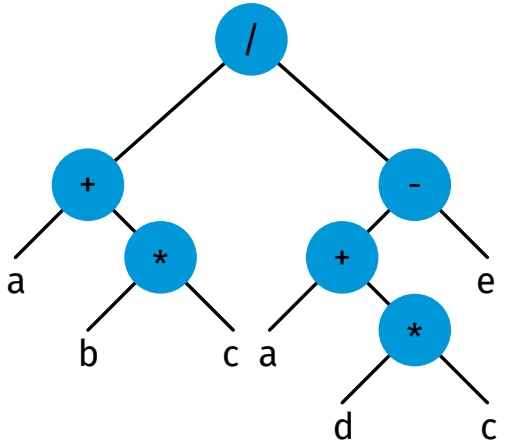
# Machine Models

## Example Stack machine model




(a+b*c)/(a+d*c-e)

Parsetree




Reverse stack

abc*+adc*+e-/

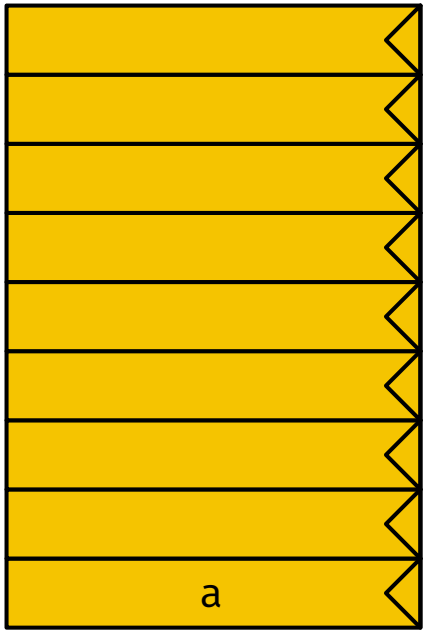
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| **push a** | **r0** | **a** |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |



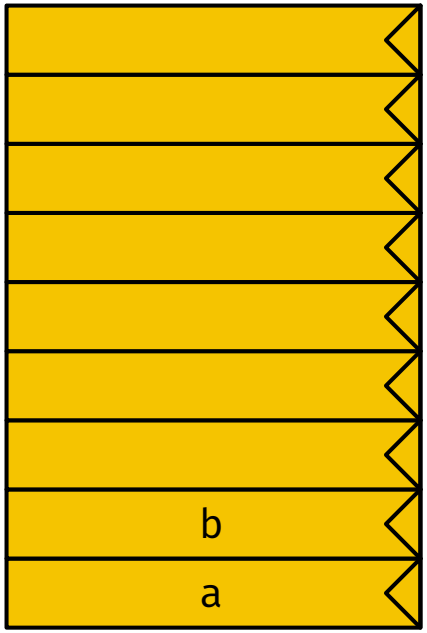
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| **push b** | **r0 r1** | **b** |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |

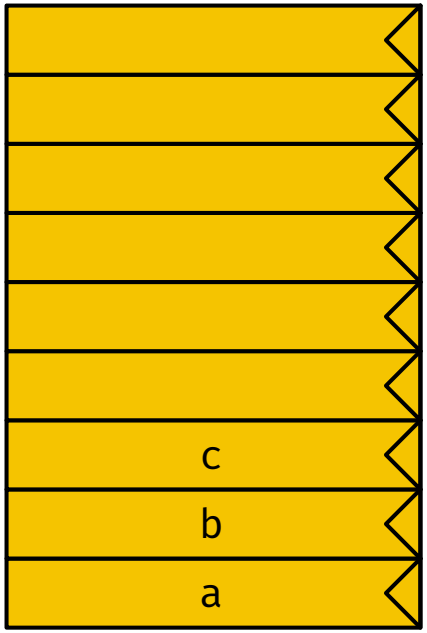


# Machine Models

## Example Stack machine model

```
                        Reverse stack
                        abc*+adc*+e-/

Program   stack         memory refs
push a    r0            a
push b    r0 r1         b
push c    r0 r1 r2      c
*         r0 r1
+         r0
push a    r0 r1         a
push d    r0 r1 r2      d
push c    r0 r1 r2 r3   c
*         r0 r1 r2
+         r0 r1
push e    r0 r1 r2      e
-         r0 r1
/         r0
```



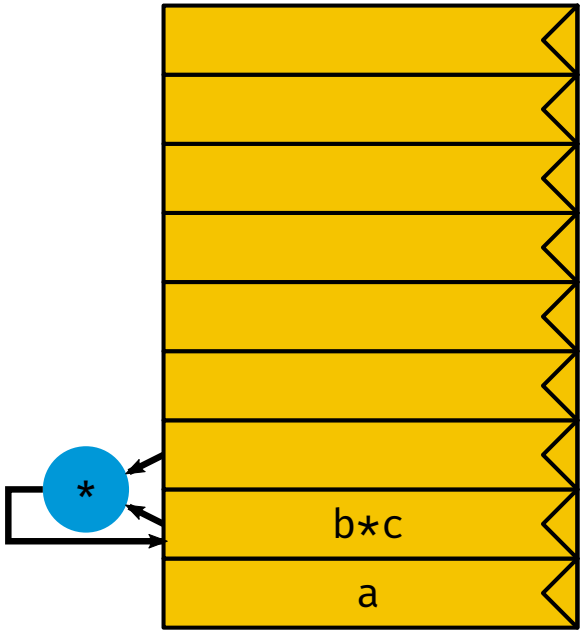
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| ***** | **r0 r1** | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |



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## Example Stack machine model

Reverse stack

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| Program | stack | memory refs |
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| push a | r0 | a |
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| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| **+** | **r0** | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
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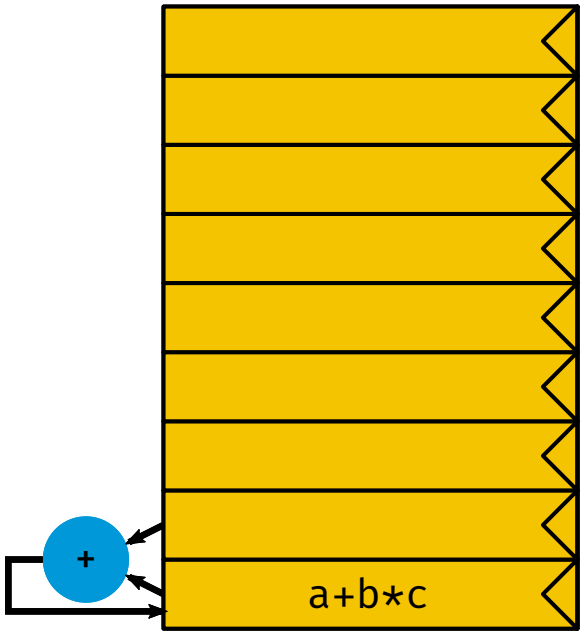


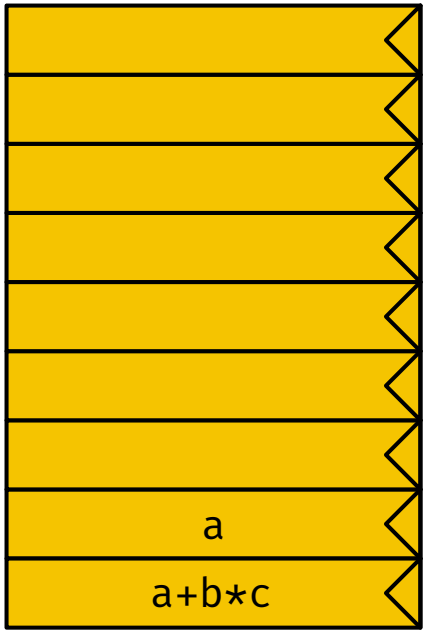
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| **push a** | **r0 r1** | **a** |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |



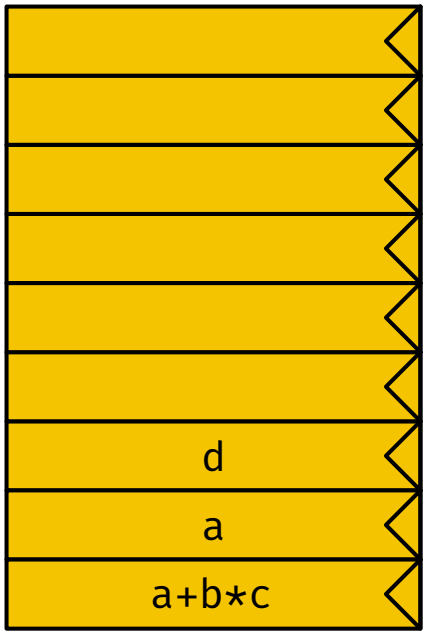
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| **push d** | **r0 r1 r2** | **d** |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |



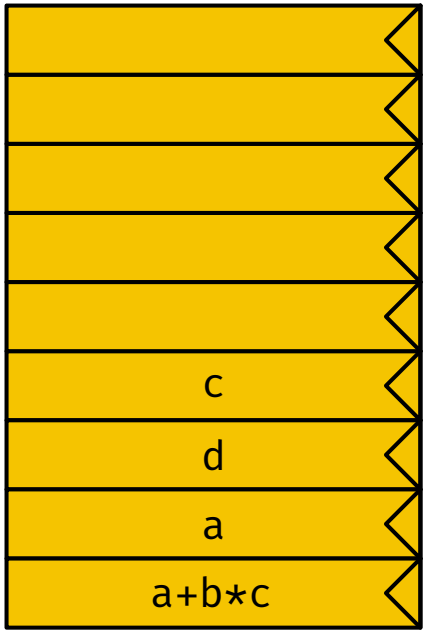
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| **push c** | **r0 r1 r2 r3** | **c** |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |



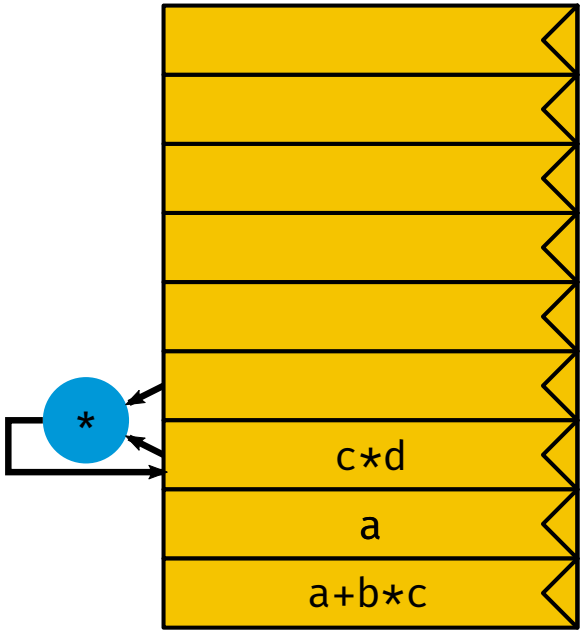
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| ***** | **r0 r1 r2** | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| / | r0 | |

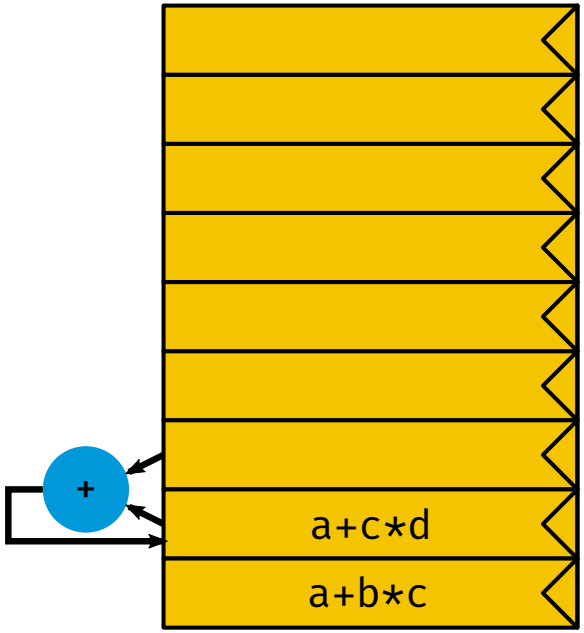


# Machine Models

## Example Stack machine model

```
                        Reverse stack
                        abc*+adc*+e-/

Program   stack         memory refs
push a    r0            a
push b    r0 r1         b
push c    r0 r1 r2      c
*         r0 r1
+         r0
push a    r0 r1         a
push d    r0 r1 r2      d
push c    r0 r1 r2 r3   c
*         r0 r1 r2
+         r0 r1
push e    r0 r1 r2      e
-         r0 r1
/         r0
```



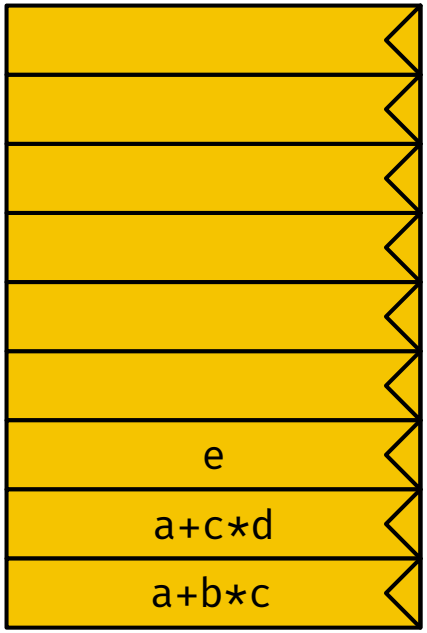
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| **push e** | **r0 r1 r2** | **e** |
| - | r0 r1 | |
| / | r0 | |



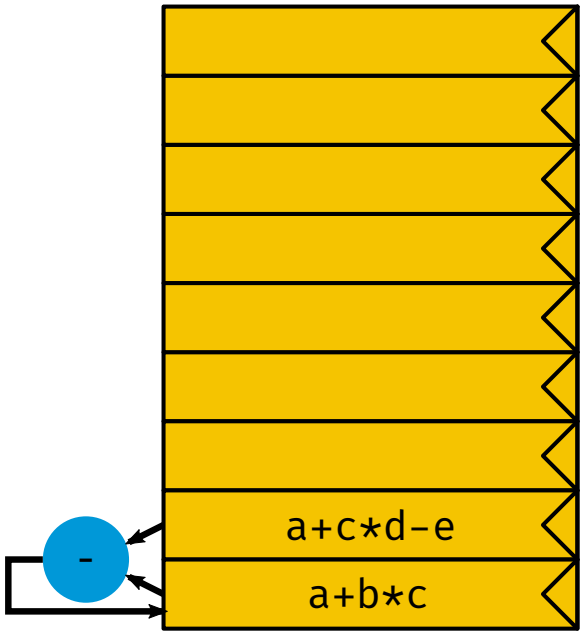
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| **-** | **r0 r1** | |
| / | r0 | |



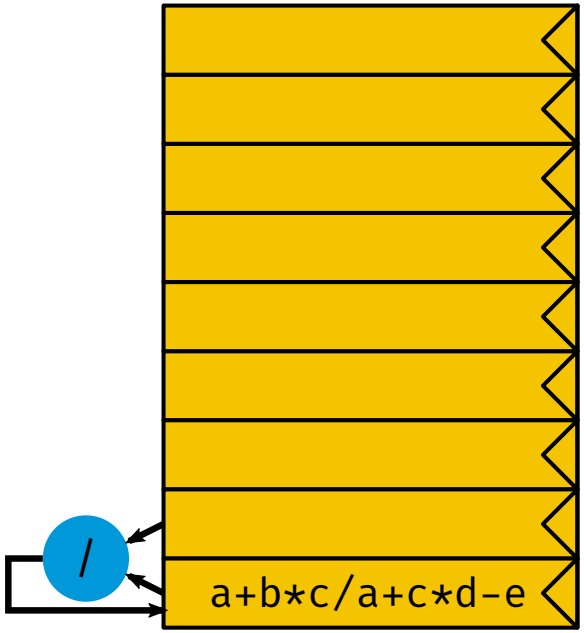
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| **/** | **r0** | |



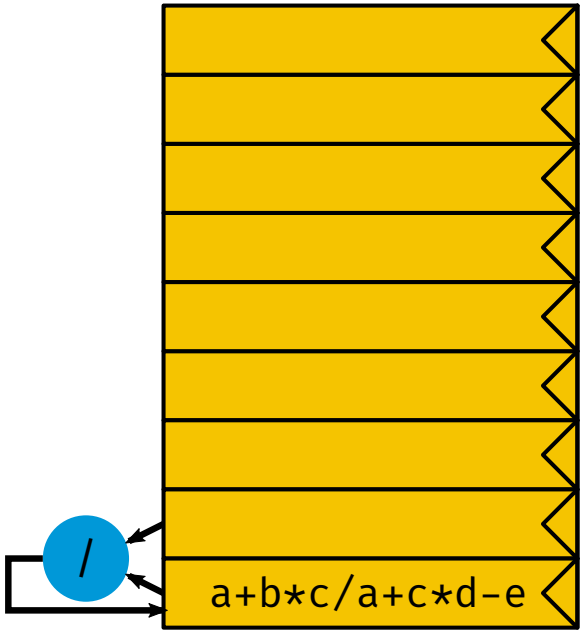
# Machine Models

## Example Stack machine model

Reverse stack

abc*+adc*+e-/

| Program | stack | memory refs |
|---|---|---|
| push a | r0 | a |
| push b | r0 r1 | b |
| push c | r0 r1 r2 | c |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | a |
| push d | r0 r1 r2 | d |
| push c | r0 r1 r2 r3 | c |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | e |
| - | r0 r1 | |
| **/** | **r0** | |



# Machine Models

## Example Stack machine model

```
                    Reverse stack
                    abc*+adc*+e-/

Program   stack        memory refs
push a    r0           a
push b    r0 r1        b
push c    r0 r1 r2     c
*         r0 r1
+         r0
push a    r0 r1        a
push d    r0 r1 r2     d
push c    r0 r1 r2 r3  c
*         r0 r1 r2
+         r0 r1
push e    r0 r1 r2     e
-         r0 r1
/         r0
```

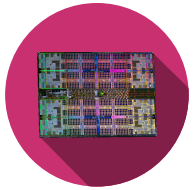
Not efficient use of registers

a & c are fetched twice

# Machine Models

## Exercise Stack machine model

Evaluate the stack in case of a stacksize of 2. How many memory references are there?



Reverse stack

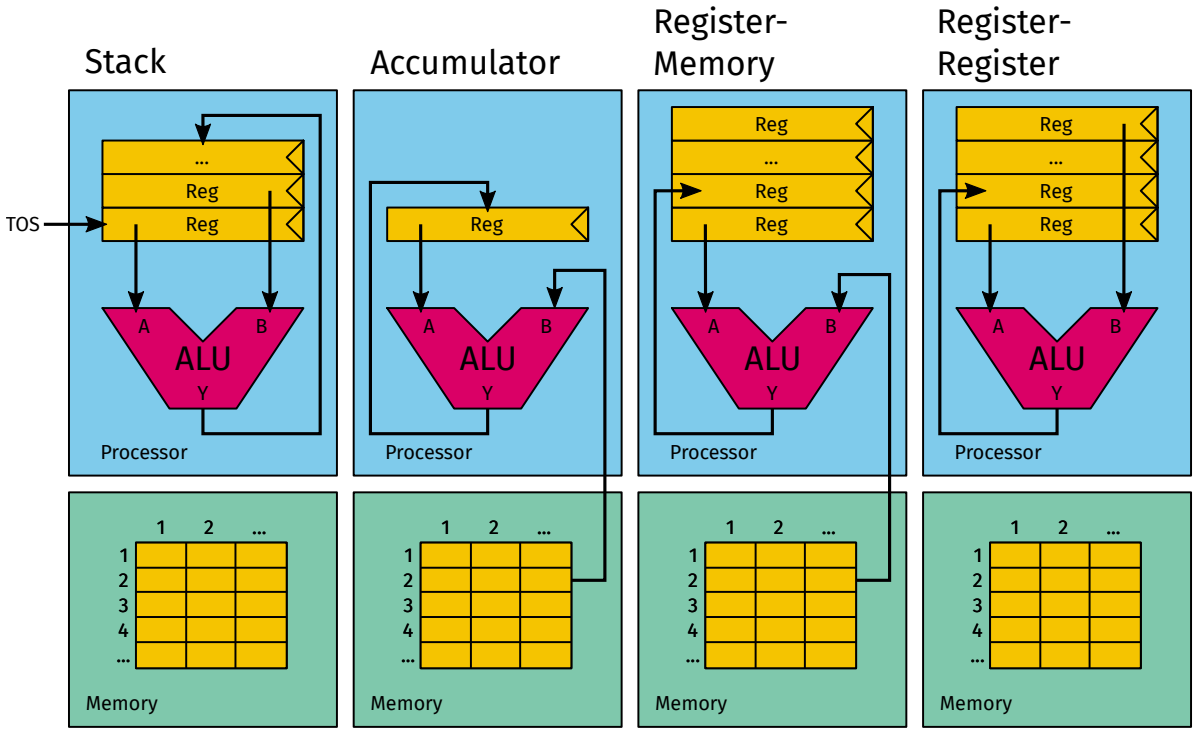
abc*+adc*+e-/

| Program | stack (size = 2) | memory refs |
|---|---|---|
| push a | r0 | |
| push b | r0 r1 | |
| push c | r0 r1 r2 | |
| * | r0 r1 | |
| + | r0 | |
| push a | r0 r1 | |
| push d | r0 r1 r2 | |
| push c | r0 r1 r2 r3 | |
| * | r0 r1 r2 | |
| + | r0 r1 | |
| push e | r0 r1 r2 | |
| - | r0 r1 | |
| / | r0 | |

# Machine Models

## Summary

c=a+b

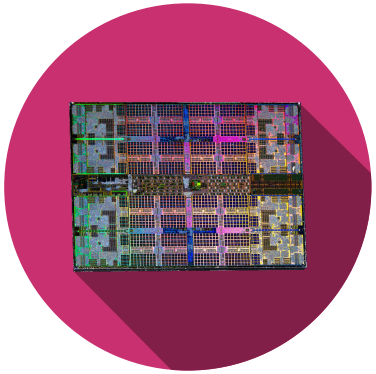



```
push a
push b
add
push c
```

```
load  a
add   b
store c
```

```
load  r1, a
add   r3, r1, b
store r3, c
```

```
load  r1, a
load  r2 ,b
add   r3, r1, r2
store r3, c
```

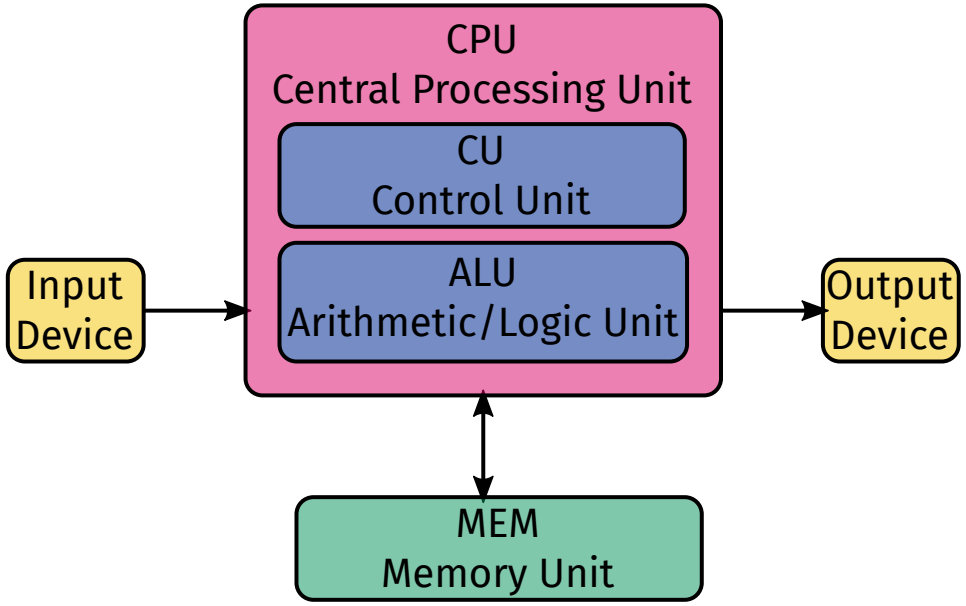


# Data Architectures

# Data Architectures

## Von Neumann

- Same memory holds data and instructions
- A single set of address/data busses between CPU and memory




**Advantages** – Cheaper, less logic, less power consumption

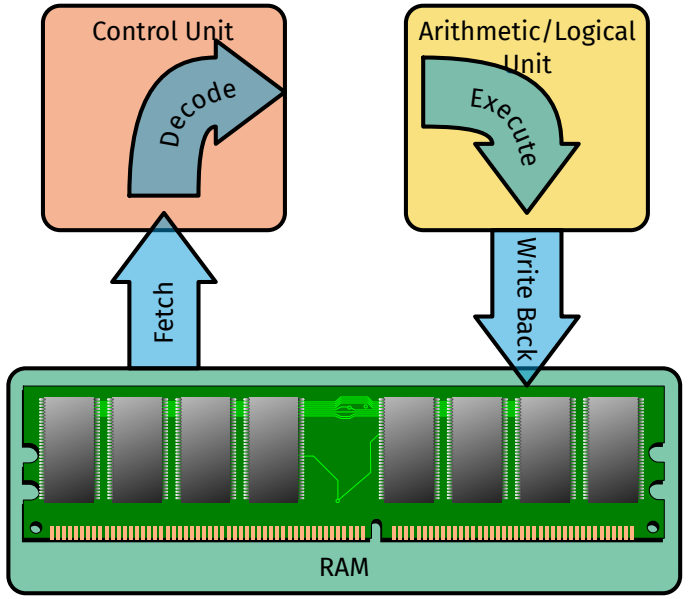
**Disadvantages** – databus speed slower than harvard, bottleneck problems

**Examples** – All Intel x86, ARM7

# Data Architectures

## Von Neumann 4-step instruction cycle

1. **Fetch** – the instruction from memory into the IR
   1. Load the MAR with the PC
   2. Place instruction into the MDR
   3. Load IR with the MDR
2. **Decode** – Examinated the IR and decides what to perform
3. **Execute** – Excutes the operation
4. **Write Back** – Write the result back into memory

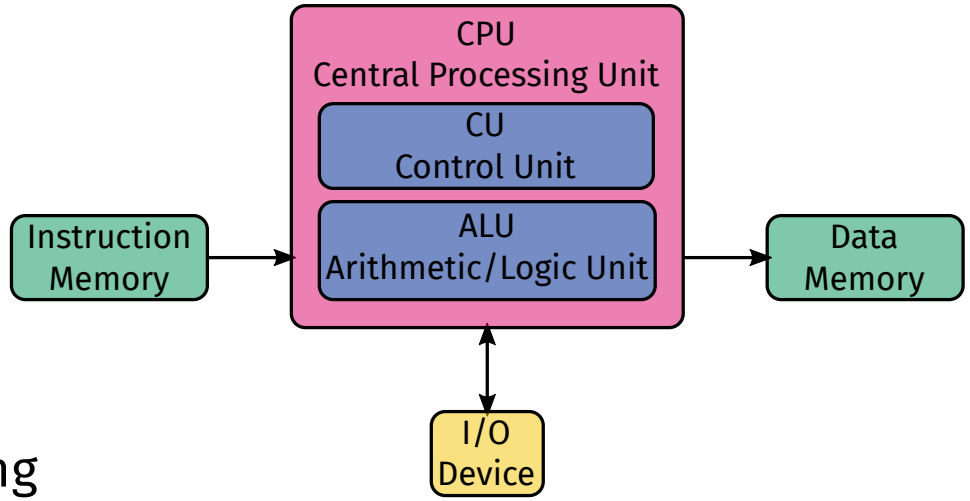



IR – Instruction Register
MAR – Memory address register
MDR – Memory data register

# Data Architectures

## Harvard

- Separate memories or data and instructions
- Two sets of address/data busses between CPU and memory




**Advantages** – greater memory bandwidth, different bit depth between instruction and data

**Disadvantages** – can't use self-modifying code

**Examples** – DSP, ARM9

# References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[2]
D. Wentzlaff, “Computer Architecture ELE 475 / COS 475.” Princeton University, 2019.

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[4]
“Difference Between Von Neumann and Harvard Architecture,” *Circuit Globe*, Jun. 10, 2021. https://circuitglobe.com/difference-between-von-neumann-and-harvard-architecture.html (accessed Nov. 03, 2022).

[5]
S. L. Harris and D. M. Harris, “Digital Design and Computer Architecture RISC-V Edition,” in *Digital Design and Computer Architecture*, First Edition., Elsevier, 2022, pp. IBC1–IBC2. doi: 10.1016/B978-0-12-820064-3.00025-8.

[6]
*TU Berlin: Course AES*, (Oct. 22, 2018). Accessed: Jul. 18, 2022. [Online Video]. Available: https://www.youtube.com/channel/UCPSsA8oxlSBjidJsSPdpjsQ

[7]
“What’s the difference between Von-Neumann and Harvard architectures?” https://www.microcontrollertips.com/difference-between-von-neumann-and-harvard-architectures/ (accessed Nov. 03, 2022).

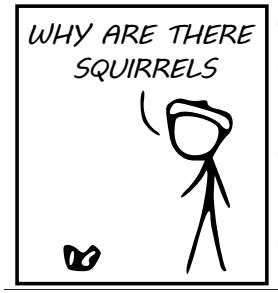
# QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

WHY AREN'T THERE DINOSAUR GHOSTS

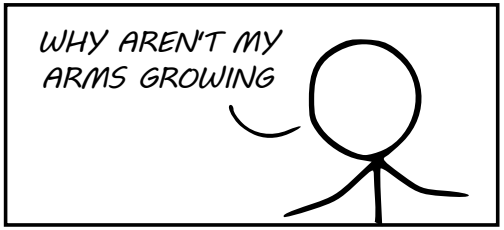
WHY IS THERE HELL IF G




WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY DO IGUANAS DIE

WHY ARE SWISS AFRAID OF DRAGONS

WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON TWITTER
WHY IS HTTPS IMPORTANT




WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD
WHY IS THERE LAVA
WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILEGRIM
WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FUNNY
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE DOGS AFRAID OF FIRE
WHY IS THERE NO KING IN EN

WHY ARE THERE ANTS IN MY LAPTOP

WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN

WHY ARE THERE FEMALE

WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE

WHY IS LIFE SO

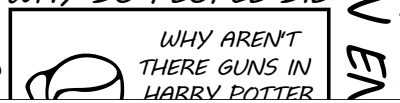
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY ARE THERE 5 GRADES

WHY

WHY 2*B | $\overline{2*B}$

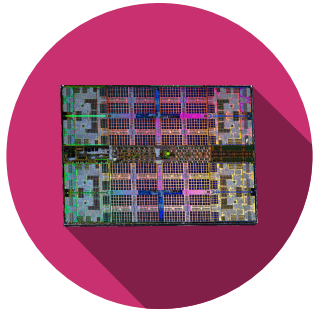
**WHAT IS https://xkcd.com/1256/**

WHY DO THEY SAY T-MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMMING MORE ACIDIC



Hes·so VALAIS WALLIS

Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



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